SAAD SHAKIR

Full Stack Developer | AI/LLM Engineer | Real-Time Applications

Multan, Pakistan • Remote (PKT, UTC+5) — available for EU & US-East overlap

saadhshakir2300@gmail.com • +92 313-6143243 • linkedin.com/in/saad-shakir-876b78338 • github.com/Saadshakir149

PROFILE

Computer Science graduate with hands-on experience in full-stack development, AI-powered applications, LLM integration, and real-time communication systems. Experienced in building web applications using React.js, JavaScript, Node.js, Python, Flask, REST APIs, and WebSockets. Worked as a Node.js Developer at SharkStack and as a Data Science / AI Research Assistant at LUMS, contributing to LLM-powered systems, speech-processing pipelines, and AI-assisted applications. Familiar with prompt engineering, embeddings, semantic retrieval, RAG concepts, tool-assisted development, and AI coding tools including Claude Code and ChatGPT. Comfortable debugging existing applications, integrating third-party APIs, deploying applications, and using AI tools to accelerate development while maintaining code quality.

TECHNICAL SKILLS

Frontend Development: React.js, JavaScript, HTML5, CSS3, Responsive Web Development, Component-Based Development

Backend Development: Node.js, Express.js, Python, Flask, REST APIs, WebSockets, Socket.IO

AI & LLM Development: OpenAI API, Groq API, LLM Integration, Prompt Engineering, Embeddings, Semantic Retrieval, RAG Concepts, Context Management, Speech-to-Text, Text-to-Speech

AI-Assisted Development: Claude Code, Claude, ChatGPT, AI-Assisted Coding, Code Generation, Debugging, Code Review Assistance

Databases: PostgreSQL, MySQL, SQL

Tools & Deployment: Git, GitHub, Postman, VS Code, Vercel, Render

Software Engineering: API Integration, Real-Time Applications, Application Testing, Debugging, Error Handling, Performance Testing, Technical Documentation

EXPERIENCE

Research assistant | Lahore University of Management Sciences (LUMS) *DEC2025 – AUG2026*

- Contributed to the development of LLM-powered applications and AI-assisted systems using Python, React.js, Flask, and LLM APIs.
- Developed and evaluated real-time AI interaction workflows involving speech-to-text, LLM processing, and text-to-speech.
- Built local speech-processing pipelines using Faster-Whisper and Pyannote for transcription and speaker diarization.
- Applied prompt engineering, language-processing techniques, and retrieval-based approaches in AI application development.
- Investigated application issues, evaluated model outputs, and tested alternative implementation approaches.
- Conducted technical research, reviewed AI/LLM documentation, and documented system architecture, experiments, and implementation workflows.
- Used AI development tools to support coding, debugging, technical research, and implementation.

Node.js Developer | Shark Stack *MAY2025 – Nov2025*

- Developed and maintained backend services and REST APIs using Node.js and JavaScript, supporting user-facing web applications and frontend integrations.
- Integrated frontend applications with backend APIs, handling request/response flows, validation, asynchronous operations, and error handling.
- Worked with React.js-based interfaces and API-driven application workflows, contributing to feature implementation and debugging.
- Investigated application issues by reproducing bugs, tracing API requests, inspecting browser behavior, and identifying implementation-level problems.
- Used Git and GitHub for version control, collaborative development, code changes, and maintaining project history.
- Tested APIs and application workflows using Postman and browser developer tools to verify functionality and troubleshoot integration issues.
- Worked within existing application structures and followed established development patterns when implementing and modifying features.

- Collaborated with team members to understand requirements, implement changes, resolve issues, and improve application reliability.

IT Support Intern | Volka Food International

Multan, Pakistan

- Provided day-to-day IT support to staff by troubleshooting computer, software, and connectivity-related issues.
- Assisted with installation, configuration, updating, and troubleshooting of software and computer systems.
- Supported basic network troubleshooting, including connectivity issues and network configuration.
- Assisted users with hardware, operating system, printer, and peripheral-related problems.
- Performed routine system checks and helped identify and resolve technical issues.
- Assisted with maintaining IT equipment and ensuring systems were available for daily operations.
- Followed technical instructions, documented issues, and communicated problems and solutions to relevant staff.

SELECTED PROJECTS

AI Assistive System — Voice-Controlled Desktop Application

Python, LLM APIs, Speech-to-Text, Text-to-Speech, Windows UI Automation

- Developed and tested a voice-controlled desktop application designed to execute computer tasks through natural-language commands.
- Created structured test scenarios for 59 voice commands, including successful execution, invalid commands, missing parameters, application-state errors, and unexpected user inputs.
- Implemented verification checks before and after actions to detect whether requested operations were actually completed.
- Tested multi-step workflows for document creation, essay generation, email composition, and application interaction.
- Investigated silent failures where the system accepted a command but failed to perform the intended action.
- Evaluated system behavior across different interaction states and documented reproducible failure conditions.

RAG-Based Document Intelligence System

Python, Embeddings, Vector Database, LLM APIs

- Built and tested a document-question-answering workflow using document processing, embeddings, semantic retrieval, and LLM-based response generation.
- Created test cases covering document upload, text extraction, chunking, retrieval, query processing, and response generation.
- Evaluated generated responses for relevance, consistency, retrieval accuracy, and unsupported answers.
- Tested edge cases including incomplete queries, irrelevant questions, missing documents, and insufficient retrieval context.

EDUCATION

BS Computer Science | Multan University of Science and Technology *2023 – 2026*

CGPA: 3.46 / 4.0

CERTIFICATIONS & LANGUAGES

- Data Science Specialization — LUMS
- Machine Learning — DataCamp
- Associate AI Engineer for Developers — DataCamp.